

Roadway Transport Data Analysis and Freight-Rate Prediction

**Initial Testing of FastAI, CatBoost, TabPFN3, LightGBM, PySR, and
TabICLv2 for Applied Prediction of Trucking Freight Rates**

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0. Preliminary Material

0.1 Module Imports

This setup loads the utilities required for system interaction, timing, data processing, symbolic regression, geospatial calculations, and hardware inspection. It also configures JuliaCall for stable, optimized multi-threaded execution before initializing PySR.

0.2 Hardware and Python, CUDA, GPU environment

Table 1: Python, CUDA, GPU Environment - Hardware Overview

Python and PyTorch	

Python version:	3.14.6
PyTorch import:	Successful
PyTorch version:	2.13.0+cu130
PyTorch CUDA runtime:	13.0
CUDA available:	True
CUDA GPU count:	1
Values reported by CUDA/PyTorch	

Python and PyTorch	

CUDA device index:	0
GPU model:	NVIDIA GeForce RTX 5060 Ti
Compute capability:	12.0
CUDA architecture:	sm_120
Total VRAM:	15.48 GiB
Streaming multiprocessors:	36
Maximum threads per block:	1024
Warp size:	32
Externally derived value	

Assumed CUDA cores per SM:	128
Derived CUDA-core count:	4608
Derivation:	36 SMs × 128 cores/SM

1.0 Initial Data Review

These initial validation steps establish the dataset's structure, confirm that each column has an appropriate data type, and identify formatting inconsistencies in text values. Detecting whitespace problems and missing data early prevents subtle grouping, matching, and modeling errors later in the pipeline.

Table 2: Initial Data - Missing Data Review

Column	Standard Missing Count	Extended Missing Count	Additional Text Missing	Missing Percent
load_id	0	0	0	0.00%
pickup	0	0	0	0.00%
delivery	0	0	0	0.00%
pickup_lat	0	0	0	0.00%
pickup_lon	0	0	0	0.00%
delivery_lat	0	0	0	0.00%
delivery_lon	0	0	0	0.00%
distance	0	0	0	0.00%
equipment	0	0	0	0.00%
weight	300	300	0	0.62%

Column	Standard Missing Count	Extended Missing Count	Additional Text Missing	Missing Percent
date	0	0	0	0.00%
market_index	374	374	0	0.78%
quote_signal	0	0	0	0.00%
posted_rate	0	0	0	0.00%

Table 3: Initial Data - Missing Weight Data Review

weight count	
missing	300
zero	0
negative	292
positive	47408

Table 4: Initial Data - Weight Data Review - Reveals Invalid Weight Values

load_id	pickup	delivery	equipment	weight	distance	date	posted_rate
68	TR-000069	Amarillo	Dayton	Reefer	-36559.0	1334.2	2025-01-2831.2701
204	TR-000205	Fresno	Tucson	Dry Van	-26670.0	496.2	2025-01-1046.9202
206	TR-000207	Columbia	Raleigh	Reefer	-20003.0	411.3	2025-01-959.1502
225	TR-000226	Greensboro	Lubbock	Reefer	-23981.0	1317.4	2025-01-2836.0002

Table 5: Initial Data - Market Index Review

market_index count	
missing	374
negative	0
zero	0
positive	47626

These analyses verify the consistency and validity of equipment, quote-signal, date, and load-identifier data, including duplicate and formatting checks. Measuring numerical correlations and categorical associations also identifies important relationships, redundant variables, and possible modeling risks.

2.0 Coordinates Review and Analysis

These steps validate the geographic data by reviewing coordinate ranges, identifying the cities represented, and mapping each city to a consistent city-center location. They also determine whether pickup and delivery cities use one or multiple coordinate pairs, helping uncover inconsistencies or synthetic location patterns.

3.0 Preprocessing

The preprocessing controller prepares the freight data for feature engineering and modeling by restoring required identifiers and coordinates, cleaning and imputing weight data, and removing obsolete or potentially leaky columns.

Preprocessing includes:

- Add missing load identifiers
- Add missing coordinate columns
- Create weight-status indicator columns
- Convert weights to absolute values
- Impute missing weights using equipment medians
- Impute missing weights using equipment-and-route medians
- Remove the original weight columns
- Remove potential target-leakage columns
- Return the preprocessed DataFrame

4.0 Feature Engineering

4.1 Derive new features

This feature-engineering controller transforms the preprocessed freight dataset into a model-ready feature set by removing target/leakage fields and adding geographic, temporal, weather, insurance, fuel-cost, and freight-market features. It also preserves posted_rate separately when present, supports both development and final-prediction inputs, and organizes key market-related columns before returning the engineered dataset.

- Load either the standard preprocessed dataset or the final-prediction dataset
- Remove posted_rate from the feature set while preserving it separately when present
- Remove any remaining leakage-prone columns
- Add pickup and delivery state features
- Identify interstate versus intrastate trips
- Add U.S. Census region features
- Add pickup and delivery timezone features
- Add daily trip-volume counts

- Add monthly trip-volume counts
- Add month-based temporal features
- Add season features
- Add weekday features
- Add holiday indicators
- Add snowfall features
- Add freezing-day weather features
- Optionally add elevation features — currently disabled
- Add insurance-risk features
- Optionally add the second insurance-risk feature set — currently disabled
- Add fuel-cost features
- Add the interaction between distance and quote_signal
- Optionally add one-year national diesel-price change — currently disabled
- Move market-related columns to the end of the dataframe
- Return the engineered dataframe and the separately preserved posted_rate data

```
DataFrame Shape: (48000, 66)
```

DataFrame Columns:		
load_id	pickup	delivery
pickup_lat	pickup_lon	delivery_lat
delivery_lon	distance	equipment
date	weight_was_zero	weight_was_missing
weight_was_negative	weight_imputed_equipment	weight_imputed_equipment_route
pickup_state	delivery_state	interstate_bool
pickup_region	delivery_region	inter_region_bool
pickup_timezone	delivery_timezone	inter_timezone_bool
day_trips_count	month_trips_count	month
season	day	day_sin
day_cos	holiday_boolean	pickup_historical_snowfall
delivery_historical_snowfall	pickup_snowfall_bool	delivery_snowfall_bool
pickup_freezing_days	delivery_freezing_days	pickup_freezing_day_bool
delivery_freezing_day_bool	pickup_altitude	delivery_altitude
pickup_insurance_risk	delivery_insurance_risk	route_insurance_risk
route_max_insurance_risk	national_diesel_price	pickup_region_diesel_price
delivery_region_diesel_price	route_average_diesel_price	diesel_change_1_week
diesel_change_4_weeks	estimated_fuel_gallons	estimated_fuel_cost
estimated_fuel_cost_per_mile	pickup_diesel_minus_national	delivery_diesel_minus_national
wti_spot_price	wti_change_5d	wti_change_20d
wti_volatility_20d	brent_spot_price	brent_wti_spread
dist_times_quote_signal	market_index	quote_signal

Figure 1: Feature Engineering - New Features Overview

4.2 Feature prediction testing for market_index and quote_signal

A range of machine-learning and symbolic-regression models was evaluated for their ability to predict missing market_index and quote_signal values. The tested approaches included FastAI, CatBoost, TabPFN 3, TabICLv2, LightGBM, and PySR, with median imputation retained as a simple baseline method. Each model was assessed to determine how effectively it could reconstruct these important freight-market features from the remaining available variables. Among the approaches tested, TabICLv2 demonstrated the strongest predictive capability for market_index and was therefore selected as the preferred method. TabICLv2 was also used to train the quote_signal prediction model, providing a consistent high-performing approach for recovering both missing market-related features.

```
#####
# TabICLv2 regression performance
Training for market_index prediction
Training: MAE=0.019571, RMSE=0.024557, R²=0.978531
Validation: MAE=0.020233, RMSE=0.025378, R²=0.977274
Testing: MAE=0.019997, RMSE=0.025097, R²=0.978141
# Generalization gaps
Validation MAE gap: 0.000662
Test MAE gap: 0.000426
Validation RMSE gap: 0.000821
Test RMSE gap: 0.000541
Test R² gap: 0.000390
#####
```

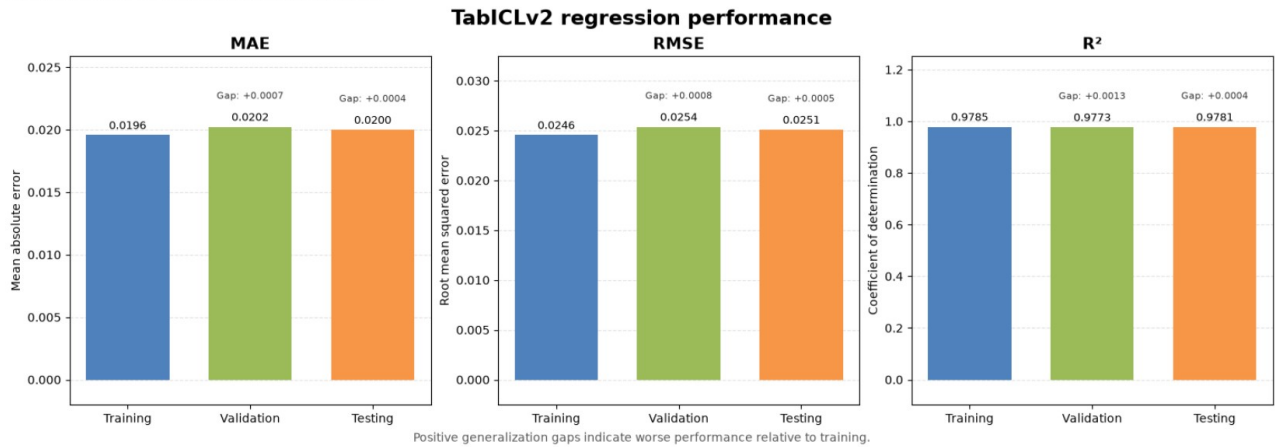


Figure 2: TabICLv2 Training - Target=market_index

```
#####
# TabICLv2 regression performance
Training for quote_signal prediction
Training: MAE=0.038850, RMSE=0.073832, R²=0.936080
Validation: MAE=0.052401, RMSE=0.083928, R²=0.918758
Testing: MAE=0.052321, RMSE=0.083824, R²=0.913831
# Generalization gaps
Validation MAE gap: 0.013552
Test MAE gap: 0.013472
Validation RMSE gap: 0.010096
Test RMSE gap: 0.009993
Test R² gap: 0.022249
#####
```

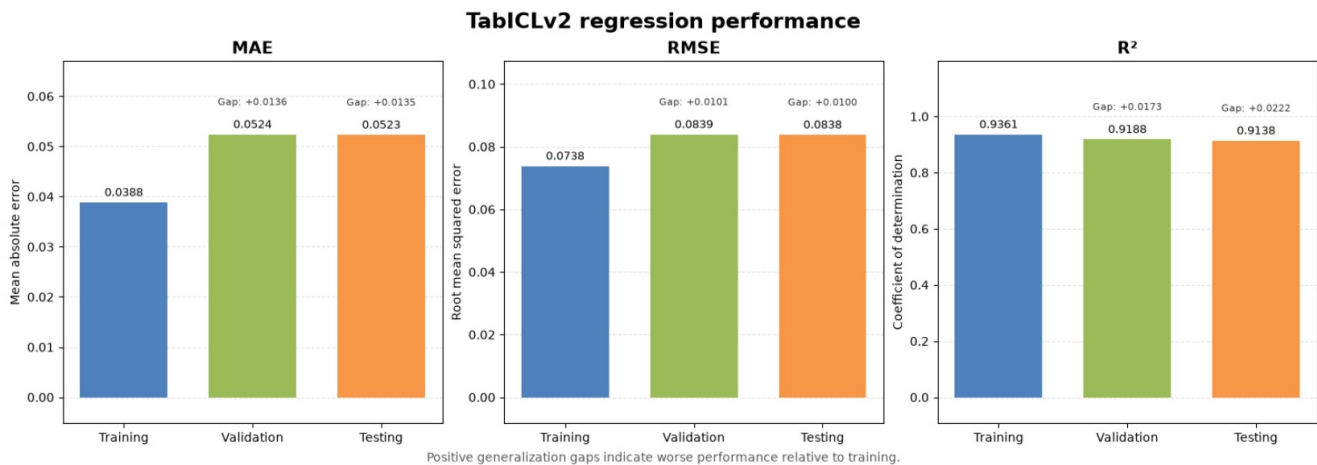


Figure 3: TabICLv2 Training - Target=quote_signal

4.3 Features review

The Features Review section performs a structured quality and validity assessment of the final engineered feature dataset before model development. It first examines the dataset shape to confirm the expected number of observations and engineered variables. The head and tail of the dataset are reviewed to verify that the feature-engineering pipeline produced sensible and consistently structured records. Column names and data types are then inspected to ensure that numerical, categorical, boolean, and temporal features are represented appropriately for downstream modeling. Missing-value analysis is performed to identify any remaining incomplete features, while a dedicated temporal leakage check verifies that features do not improperly incorporate information from the future. Finally, correlation and association analyses are conducted across all features to understand their relationships, identify potentially redundant variables, and assess their likely relevance for freight-rate prediction.

5.0 Predict predicted_rate

5.1 TabICLv2 model training

TabICLv2 was found to have the best predictive capacity with this data. For that reason it is here trained with train-test.csv, specifically for the purpose of predicting posted_rate, and predicted_rate in the testing/validation data sets, validation.csv and december-chart-inputs.csv.

```
#####
# TabICLv2 regression performance
Training for posted_rate prediction
Training:  MAE=52.014249, RMSE=584.208932, R²=0.846224
Validation: MAE=81.186615, RMSE=596.954075, R²=0.837712
Testing:    MAE=75.182604, RMSE=551.361840, R²=0.860776
# Generalization gaps
Validation MAE gap:  29.172365
Test MAE gap:       23.168355
Validation RMSE gap: 12.745143
Test RMSE gap:      -32.847092
Test R² gap:        -0.014552
#####
```

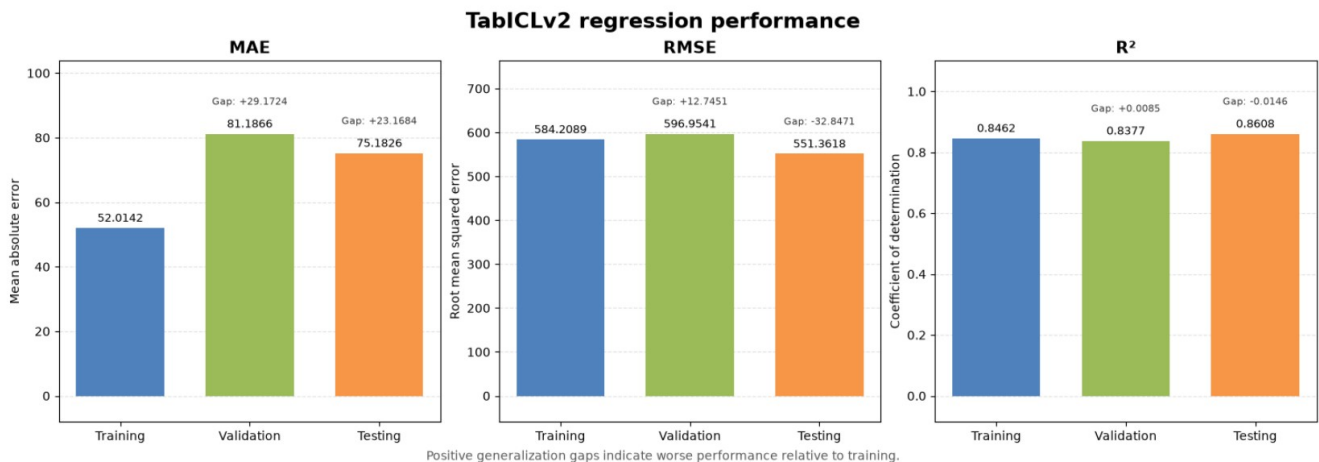


Figure 4: TabICLv2 Training - Target=posted_rate

5.2 Apply TabICLv2 to validation.csv

The now ready preprocessing, feature engineering, and trained TabICLv2 model was applied to the validation.csv data. This validation.csv data set is not missing market_index and quote_signal columns/features.

5.2.2 Prediction results for validation.csv

```
import Utils.final_prediction as fp
import importlib
fp = importlib.reload(fp)
fp.fill_validation_predicted_values()
```

Filled and saved 12,000 predicted_rate values.

	load_id	predicted_rate
0	TE-000001	835.577209
1	TE-000002	5026.601074
2	TE-000003	5214.097656
3	TE-000004	4095.393555
4	TE-000005	1822.592407
...	...	...
11995	TE-011996	910.366516
11996	TE-011997	1645.984985
11997	TE-011998	983.428223
11998	TE-011999	3426.381348
11999	TE-012000	634.896118

12000 rows × 2 columns

Figure 5: Prediction of predicted_rate for validation.csv

5.3 Apply TabICLv2 to december-chart-inputs.csv

For december-chart-inputs.csv, prediction again starts with preprocessing, and feature engineering. The december-chart-inputs.csv data set is quite different than the validation.csv data set, and it is missing the market-index and quote_signal features/columns. TabICLv2 was specifically prepared and trained to predict these 2 columns if missing. So TabICLv2 was used to create and add, by prediction, the columns market_index and quote_signal. Then another TabICLv2 that was specifically trained to predict predicted_rate was applied to arrive at the new, predicted predicted_rate.

5.3.3 Prediction results for december-chart-inputs.csv

```
import Utils.final_prediction as fp
import importlib
fp = importlib.reload(fp)

fp.fill_december_chart_predicted_values()
```

Filled and saved 31 predicted_rate values.

	pickup	delivery	distance	equipment	weight	date	predicted_rate
0	Lexington	Fort Wayne	360	Dry Van	32000	2025-12-01	794.931763
1	Lexington	Fort Wayne	360	Dry Van	32000	2025-12-02	803.842834
2	Lexington	Fort Wayne	360	Dry Van	32000	2025-12-03	796.750366
3	Lexington	Fort Wayne	360	Dry Van	32000	2025-12-04	804.206604
4	Lexington	Fort Wayne	360	Dry Van	32000	2025-12-05	809.844238
5	Lexington	Fort Wayne	360	Dry Van	32000	2025-12-06	806.661743
6	Lexington	Fort Wayne	360	Dry Van	32000	2025-12-07	808.752991
7	Lexington	Fort Wayne	360	Dry Van	32000	2025-12-08	798.296204
8	Lexington	Fort Wayne	360	Dry Van	32000	2025-12-09	806.570801
9	Lexington	Fort Wayne	360	Dry Van	32000	2025-12-10	804.115662
10	Lexington	Fort Wayne	360	Dry Van	32000	2025-12-11	817.300293

Figure 6: Prediction of predicted_rate for december-chart-inputs.csv

6.0 Conclusion

The roadway freight-rate analysis began with a 48,000-record dataset that required preprocessing and feature engineering before it could support reliable predictive modeling. Two important market-related columns, market_index and quote_signal, contained missing values, so the modeling workflow first focused on reconstructing these variables from the remaining information rather than simply applying generic imputation. Feature engineering expanded the dataset to 66 columns, adding geographic, temporal, weather, insurance-risk, fuel-price, crude-oil, route-cost, and operational features intended to provide a broader representation of the conditions influencing trucking rates.

The engineered fuel and energy features were particularly relevant because quote_signal exhibited meaningful association with fuel-price conditions, making diesel and crude-oil variables useful contextual predictors. TabICLv2 was evaluated alongside several alternative tabular and symbolic-

regression approaches and demonstrated the strongest overall predictive capability for the missing market variables. It was therefore trained and used to estimate `market_index` and `quote_signal`, allowing these features to be incorporated into the completed modeling dataset and subsequently used in freight-rate prediction.

For `market_index`, TabICLv2 achieved exceptionally strong performance, with R^2 values of approximately 0.979 for training, 0.977 for validation, and 0.978 for testing, indicating that nearly 98% of the observed variance was captured consistently across the data splits. Prediction of `quote_signal` was also strong, producing R^2 values of approximately 0.936 for training, 0.919 for validation, and 0.914 for testing, although the larger generalization gap indicates that this variable is inherently more difficult to reconstruct than `market_index`. Finally, when TabICLv2 was applied to the primary `posted_rate` prediction task, it achieved R^2 values of approximately 0.846 for training, 0.838 for validation, and 0.861 for testing.

Overall, the results indicate that the feature-engineering and model-development pipeline was able to recover the two initially incomplete market variables with substantial predictive accuracy and then use the completed feature set to explain roughly 84–86% of the variance in roadway freight rates. The close agreement between training, validation, and testing performance also suggests that the resulting TabICLv2 models generalize reasonably well rather than relying primarily on training-set-specific relationships. These results provide a strong initial foundation for further model refinement, residual and outlier analysis, feature selection, and potential ensemble approaches aimed at improving prediction accuracy for the comparatively small number of freight movements that remain difficult to model.

TabICLv2 Freight-Rate Predictions Recap	
Prediction Target	Testing R^2
<code>market_index</code>	0.978141
<code>quote_signal</code>	0.913831
<code>posted_rate</code>	0.860776

Figure 7: TavICLv2 Freight-Rate Predictions Recap

6.0 Room to improve, next steps

The current results provide a strong baseline, but there very much is substantial room to improve freight-rate prediction! This really just scratches the surface. By expanding the feature space and

testing more adaptive modeling approaches. The next stage should focus on richer details, especially where current engineered variables may only approximate the underlying cost and demand conditions affecting a load.

- **Improve fuel features** with more accurate, localized, and relevant fuel price data.
- **Engineer features from GPS and sensor data.**
- **Integrate vehicle telemetry** such as speed, idle time, fuel efficiency, engine load, braking, and utilization measures.
- **Add commodity futures** and related market data to better capture current and forward-looking costs.
- **Test reinforcement-learning approaches** for dynamic freight-rate estimation!